MATH 261, Homework Set 8 Name: _____

1. Let $\mathbf{u}$ and $\mathbf{v}$ be two vectors. Give the geometric meaning of the magnitude of the cross product $|\mathbf{u} \times \mathbf{v}|$.

1. Final answer

2. Parameterize the following each surface:

(a) The portion of the plane $z = 10$ which is above the square $-1 \leq x \leq 1$, $-2 \leq y \leq 2$.

2(a) Final answer

(b) The portion of the plane $x - y + z = 4$ that is within the cylinder $x^2 + y^2 = 9$.

2(b) Final answer

2. (continued)

(c) The paraboloid $z = 2x^2 + y^2$ that is within the cylinder $x^2 + y^2 = 1$.

2(c) Final answer

(d) The paraboloid $z = x^2 + y^2$ that is above the triangle with vertices at $(0, 0)$, $(0, 1)$ and $(1, 1)$.

2(d) Final answer

(e) The portion of the cone $z = \sqrt{x^2 + y^2}$ that lies between the cylinders $x^2 + y^2 = 4$, $x^2 + y^2 = 16$.

2(e) Final answer

(f) The portion of cylinder $x^2 + y^2 = 1$ that lies between the planes $z = 0$ and $z = 3$ in the first octant.

2(f) Final answer

3. Consider the surface $\mathbf{r}(u, v) = \langle u,\, v^3,\, uv \rangle$, for $0 \leq u \leq 1$, $-2 \leq v \leq 2$. Suppose that this surface has density $\rho(u, v) = u^2 + v^2$. Set up, but do not solve, an integral which represents the mass of this surface.

3. Mass integral

4. Consider the surface $\mathbf{r}(u, v) = \langle u + v,\, u - v,\, u^2 + v^2 \rangle$, for $0 \leq u \leq 1$, $0 \leq v \leq 1$. Suppose that this surface has density $\rho(u, v) = e^{u+v}$. Set up, but do not solve, an integral which represents the mass of this surface.

4. Mass integral

5. Consider the surface formed by rotating the curve $y = x^2$, for $0 \leq x \leq 2$, around the x-axis.

(a) Parameterize this surface.

5(a) Final answer

(b) Set up, but do not solve, an integral which represents the area of this surface.

5(b) Area integral

6. Consider the surface given by $z = 1 + x^2 + y^4$ which is above the square $0 \leq x \leq 1$, $0 \leq y \leq 1$.

(a) Parameterize this surface.

6(a) Final answer

(b) Set up, but do not solve, an integral which represents the area of this surface.

6(b) Area integral

7. Consider the surface formed by rotating the curve $y = e^x$, for $0 \leq x \leq 1$, around the y-axis.

(a) Parameterize this surface.

7(a) Final answer

(b) Set up, but do not solve, an integral which represents the area of this surface.

7(b) Area integral

8. Consider the surface consisting of the portion of the cylinder $x^2 + y^2 = 4$, which is above the plane $z = 0$ and below the plane $x + y + 2z = 4$.

(a) Parameterize this surface.

8(a) Final answer

(b) Set up, but do not solve, an integral which represents the area of this surface.

8(b) Area integral

9. Consider the surface formed by rotating the curve $y = 1/\sqrt{x}$, for $1 \leq x \leq 4$, around the x-axis.

(a) Parameterize this surface.

9(a) Final answer

(b) Set up, but do not solve, an integral which represents the area of this surface.

9(b) Area integral

10. Consider the surface $\sqrt{x^2+y^2} = z$, for $1 \leq z \leq 2$.

(a) Parameterize this surface.

10(a) Final answer

(b) Set up and solve an integral which represents the area of this surface.

10(b) Area integral	**10(b) Area**

(c) Define $f(x,y,z) = \dfrac{1}{\sqrt{x^2+y^2}}$. Compute the surface integral $\iint_S f \, d\sigma$.

10(c) Surface integral

11. Consider the surface $x^2 + y^2 + z^2 = 1$.

(a) Parameterize this surface.

11(a) Final answer

(b) Set up and solve an integral which represents the area of this surface.

11(b) Area integral

11(b) Area

(c) Define $f(x, y, z) = z^2$. Compute the surface integral $\iint_S f \, d\sigma$.

11(c) Surface integral

12. Consider the surface consisting of the portion of the plane $x + 2y + 3z = 6$ which lies in the first octant.

(a) Parameterize this surface.

12(a) Final answer

(b) Set up and solve an integral which represents the area of this surface.

12(b) Area integral

12(b) Area

(c) Define $f(x, y, z) = z$. Compute the surface integral $\iint_S f \, d\sigma$.

12(c) Surface integral

13. Consider the surface $x^2 + y^2 = 1$, for $1 \leq z \leq 5$.

(a) Parameterize this surface.

13(a) Final answer

(b) Set up and solve an integral which represents the area of this surface.

13(b) Area integral

13(b) Area

(c) Define $f(x, y, z) = z^2$. Compute the surface integral $\iint_S f \, d\sigma$.

13(c) Surface integral

14. Consider the surface $x^2 + y^2 + z^2 = 4$ above the plane $z = \sqrt{2}$.

(a) Parameterize this surface.

14(a) Final answer

(b) Set up and solve an integral which represents the area of this surface.

14(b) Area integral	**14(b) Area**

(c) Define $f(x, y, z) = 10$. Compute the surface integral $\iint_S f \, d\sigma$.

14(c) Surface integral